Software Requirements Specification

Multilingual AI Audiobook Generator

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Chapter 1

Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the complete functional and non-functional requirements of the **Multilingual AI Audiobook Generator**. This document serves as a formal agreement between stakeholders and developers regarding system behavior and constraints.

1.2 Scope

The Multilingual AI Audiobook Generator is a web-based application that converts textual documents into audiobook-style MP3 narration. The system supports automatic language detection, multilingual translation, chapter-based processing, and downloadable audio output.

1.3 Intended Audience

This document is intended for:

- Students and academic evaluators
- Project supervisors
- Developers and maintainers
- End users

1.4 Definitions and Acronyms

Term	**Description**
SRS	Software Requirements Specification
NLP	Natural Language Processing
TTS	Text-to-Speech
API	Application Programming Interface
MP3	Audio file format

Chapter 2

Overall Description

2.1 Product Perspective

The system is a standalone web application developed using Python and Streamlit. It integrates third-party AI services for natural language processing and speech synthesis.

2.2 Product Functions

- Upload documents in PDF, DOCX, or TXT format
- Extract and segment text into chapters
- Detect the original language automatically
- Translate and rephrase content
- Generate audiobook-style MP3 files
- Allow playback and download of audio

2.3 User Classes and Characteristics

User Type	Characteristics
General User	Basic computer knowledge
Student	Uses system for learning materials
Visually Impaired User	Relies on audio output
Admin	Maintains system and APIs

2.4 Operating Environment

- Operating System: Windows, macOS, Linux
- Programming Language: Python
- Deployment Platform: Streamlit Cloud
- Browser: Chrome, Edge, Firefox

2.5 Design and Implementation Constraints

- Internet connectivity required
- Dependency on third-party APIs
- API usage limits may apply

2.6 Assumptions and Dependencies

- Users upload readable documents
- API keys are configured correctly
- Server resources are sufficient

Chapter 3

System Features

3.1 Document Upload Feature

The system provides a file upload interface allowing users to upload PDF, DOCX, or TXT files.

3.2 Language Detection Feature

The system automatically detects the source language using AI-based NLP techniques.

3.3 Translation and Rephrasing Feature

The system translates content into a selected language and rewrites it in an audiobook-friendly style.

3.4 Audio Generation Feature

The system converts processed text into MP3 audio using text-to-speech technology.

Chapter 4

Functional Requirements

4.1 FR-1 File Upload

The system shall allow users to upload documents in supported formats.

4.2 FR-2 Text Extraction

The system shall extract textual content accurately from uploaded documents.

4.3 FR-3 Language Detection

The system shall detect the original document language automatically.

4.4 FR-4 Translation

The system shall translate content into the selected target language.

4.5 FR-5 Audio Output

The system shall generate MP3 audio files for original and translated text.

4.6 FR-6 Download Facility

The system shall allow users to download generated audio files.

Chapter 5

Non-Functional Requirements

5.1 Performance Requirements

The system should process documents and generate audio within acceptable time limits.

5.2 Usability Requirements

The interface should be simple, intuitive, and user-friendly.

5.3 Security Requirements

API keys must be protected and sensitive data must not be exposed.

5.4 Reliability Requirements

The system should handle errors gracefully and display appropriate messages.

5.5 Portability Requirements

The application should run on multiple operating systems without modification.

5.6 Scalability Requirements

The system should handle multiple users without significant performance degradation.

Chapter 6

System Architecture

- The Multilingual AI Audiobook Generator follows a layered system architecture.
- The system is divided into four major layers:
  - User Interface Layer
  - Processing Layer
  - AI & Language Processing Layer
  - Output Layer
- The User Interface Layer is developed using Streamlit and provides functionalities such as:
  - Uploading PDF, DOCX, and TXT documents
  - Selecting target language
  - Choosing audio generation options
- The Processing Layer is responsible for extracting textual content from uploaded documents.
- This layer supports document parsing for multiple file formats and prepares clean text for further processing.
- The AI & Language Processing Layer uses Google Gemini for:
  - Automatic language detection
  - Translation into selected target language
  - Rephrasing content into an audiobook-friendly style
- The Text-to-Speech Layer converts processed text into audio using the gTTS library.
- The Output Layer stores generated MP3 files and provides options for:
  - Audio playback within the browser
  - Downloading MP3 files
- The modular architecture allows easy scalability and future enhancements.

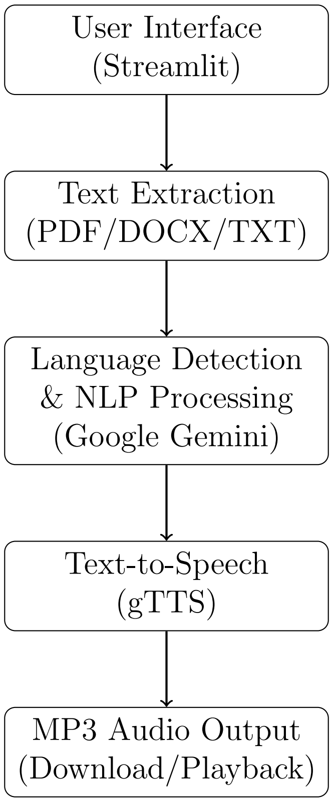

Figure 6.1: System Architecture of Multilingual AI Audiobook Generator

6.1 Use Case Diagram

- A Use Case Diagram represents the interaction between users and the system.
- The primary actor of the system is the **User**.
- The User interacts with the system to convert textual documents into audiobook-style MP3 files.
- The system supports document upload, language detection, translation, and audio generation functionalities.
- The Use Case Diagram helps in understanding the functional scope of the system.
- It clearly defines what actions a user can perform and how the system responds.

6.1.1 Actors

- **User**
  - Uploads documents
  - Selects target language
  - Generates audiobook audio
  - Downloads MP3 files

6.1.2 Use Cases

- Upload Document
- Extract Text
- Detect Language
- Translate and Rephrase Text
- Generate Original Language Audio
- Generate Translated Audio
- Play Audio
- Download MP3 File

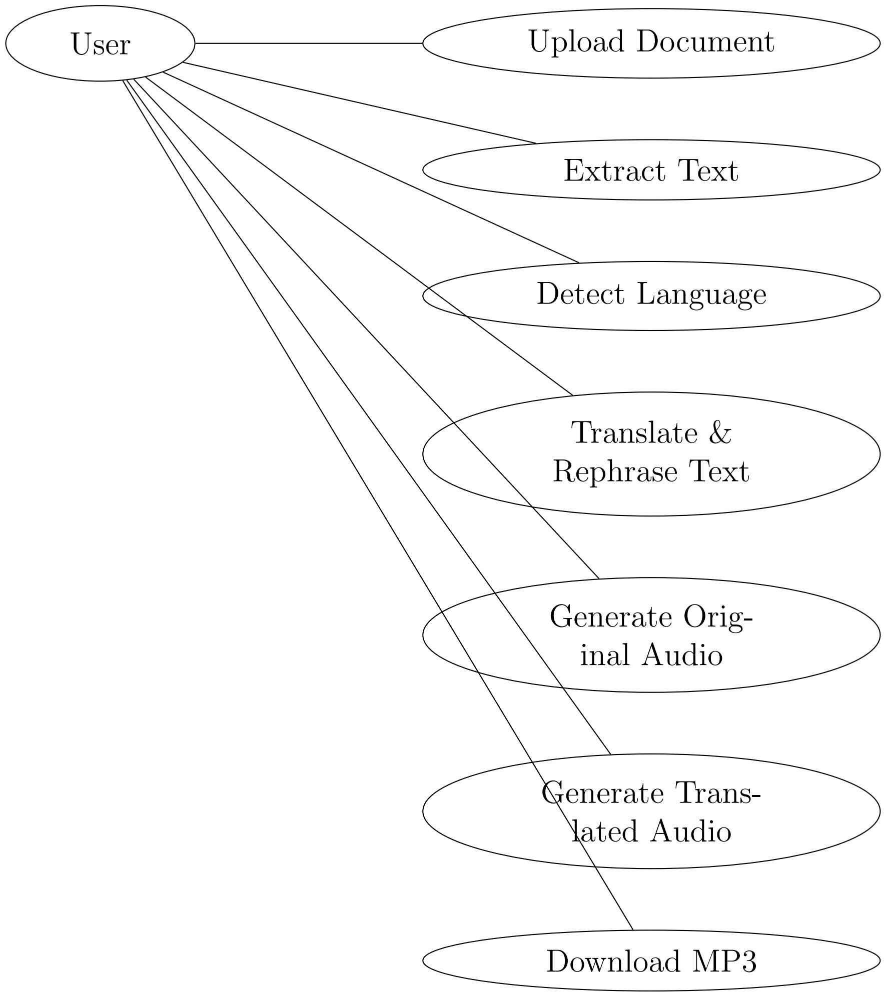

Figure 6.2: Use Case Diagram for Multilingual AI Audiobook Generator

- The User initiates the process by uploading a document.
- The system extracts text from the uploaded document.
- The system automatically detects the original language of the text.
- The extracted text can be translated and rephrased into a selected target language.
- The system generates audiobook audio in original and/or translated language.
- The generated MP3 files can be played or downloaded by the user.

Chapter 7

Future Enhancements

- Voice and accent selection
- Mobile application support
- Offline audiobook generation
- Cloud storage integration
- Support for more languages

Chapter 8

Conclusion

This document presented a comprehensive Software Requirements Specification for the Multilingual AI Audiobook Generator. The system aims to improve accessibility and learning through AI-powered audiobook generation.

References

- IEEE Std 830-1998 – Software Requirements Specification
- IEEE Std 29148-2018 – Requirements Engineering
- Streamlit Documentation – https://docs.streamlit.io